MASTERING SPORTS & NUTRITION: THE ULTIMATE GUIDE FOR STUDENTS

@ThunderStudy

Comprehensive guide on nutrition, its components, and its crucial role in sports performance for Class 12 Physical Education.

1. The Foundations of Fuel

WHAT IS FOOD?
Any substance consumed for survival. Basic need to stay alive.

WHAT IS NUTRITION?
Dynamic process of consuming food to nourish the body, build tissues, provide energy, and stay healthy.

WHAT IS A BALANCED DIET?
Diet containing all essential food groups in correct proportions.

THE 6 PILLARS OF A BALANCED DIET
- Carbohydrates
- Proteins
- Fats
- Vitamins
- Minernis
- Water

2. The Building Blocks: Macro & Micro Nutrients

MACRONUTRIENTS: THE MAJOR PLAYERS

CARBOHYDRATES (Carbs)
Nutrients required in large amounts. They are the main part of your food.

PROTEIN
Building and repairing body tissues, muscles, nalls, halr, cells.

FAT
Reserve energy, protects organs, regulates temp, helps absorb vitamins A, D, E, & K.

WATER
Transports nutrients, regulates temp, chemical reactions, Recommended 4-5 liters/day.

MICRONUTRIENTS: THE SUPPORTING CAST

VITAMINS
Nutrients required in small amounts, but vital for supporting the functions of macronutrients and overall health.

MINERALS
Inorganic elements, essential for body functions (e.g., Bone health: Calcium, Blood formation: Iron).

VITAMIN TYPES & DEFICIENCIES

Type	Description	Members	Mnemonic
FAT-SOLUBLE	Dissolve in fat, stored.	A, D, K, K	(Remaining)
WATER-SOLUBLE	Dissolve in water, not stored.	B-complex, C	"BC pani-pani ho gaya"

Vitamin	Scientific Name	Deficiency Disease
Vitamin A	Retinol	Night Blindness
Vitamin B1	Thiamine	Beriberl
Vitamin B3	Niacin	Pellagra
Vitamin B6	Pyridoxine	Anemia
Vitamin B7	Biotin	Dermatitis (Skin issues)
Vitamin C	Ascorbic Acid	Scurvy
Vitamin D	Cololferol	Rickets
Vitamin K	Phylloquinone	Excessive bleeding (no clotting)

3. Nutritive vs. Non-Nutritive Components

NUTRITIVE COMPONENTS
Provide energy or colories. Includes carbs, proteins, fats, vitamins, and minerals.

NON-NUTRITIVE COMPONENTS
Do not provide energy, but important functions, Includes fiber, water, color & flavor compounds.

THE POWER OF FIBER (BOUGHEGS): Crucial for digestion. Lack can lead to constipation.

4. Your Guide to a Healthy Weight

WHAT IS A HEALTHY WEIGHT?
Ideal weight for height; fit body, good posture, disease protection.

CALCULATE YOUR BMI

$$BMI = \frac{\text{WEIGHT (in kg)}}{[\text{HEIGHT (in m)}]^2}$$

UNDERSTANDING YOUR BMI SCORE

<16.5	18.5-24.9	25.0-29.9	≥30.0
(Underweight)	Healthy Weight	(Overweight)	Obese

THE PITFALLS OF DIETING
Extreme calorie restriction is harmful. Loads to weakness, slower metabolism, nutrient deficiencies. Skipping meals is not the solution.

5. Food Myths Busted

Myth	Fact
MYTH: Carbohydrates make you fat.	**FACT:** Carbs are primary energy source. Only excessive consumption leads to fat storage.
MYTH: Eggs raise cholesterol.	**FACT:** Eggs are high-quality protein, in moderation, they don't significantly raise harmful cholesterol for most.
MYTH: Drinking water during meals makes you gain weight.	**FACT:** Water has zero calories and cannot cause fot gain. It is a non-nutritive component.

6. Fueling for Victory: A Sports Diet Plan

2-3 hours before — PRE-COMPETITION DIET
Focus on high-carb diet for sastained energy. Consume easily digestible carbs, ensure good hydration.

Short breaks — DURING COMPETITION DIET
Replenish fluids, get instant energy. Bananas, glucose water, or carb-rich drinks.

1-2 hours after — POST-COMPETITION DIET
Focus on recovery. Consume protein-rich meal with carbs to repair muscle and restore energy.